Freight Rate Prediction Challenge

Machine Learning Engineer Assessment — Technical Report

Objective

Predict posted freight rates for the 12,000 unlabeled validation loads and provide a daily December 2025 prediction series for the supplied fixed route.

Data exploration and quality checks

The labeled development set contains 48,000 rows from January 1 through October 31, 2025, with 14 columns including the target **posted_rate**. The validation set contains 12,000 rows. In the labeled data, 300 weight values are missing and 292 weights are negative; validation contains 165 missing and 145 negative weights. Missing weights are imputed with the median calculated from the applicable training data. Negative weights are retained as supplied in this baseline and should be investigated as a data-quality issue in future iterations. The model uses pickup, delivery, distance, equipment, weight, and date-derived features; identifiers, coordinates, market_index, and quote_signal are excluded to keep the feature schema compatible with the fixed December inputs.

Feature engineering and model choice

A **CatBoostRegressor** was selected because it supports categorical features such as pickup, delivery, and equipment directly and can model nonlinear relationships. The model uses route cities, distance, equipment, weight, and date-derived features (month, day of week, day of month, day of year, and weekend flag). The target is transformed with *log1p* during training and converted back with *expm1* for predictions, with predictions constrained to positive values.

Validation and split strategy

A chronological holdout was used to mimic forecasting a later period: records dated before October 1, 2025 formed the training set (43,147 rows), and October 2025 formed the holdout set (4,853 rows). Early stopping selected 466 iterations. The model was then retrained on all 48,000 labeled records with this iteration count before generating the final validation and December predictions.

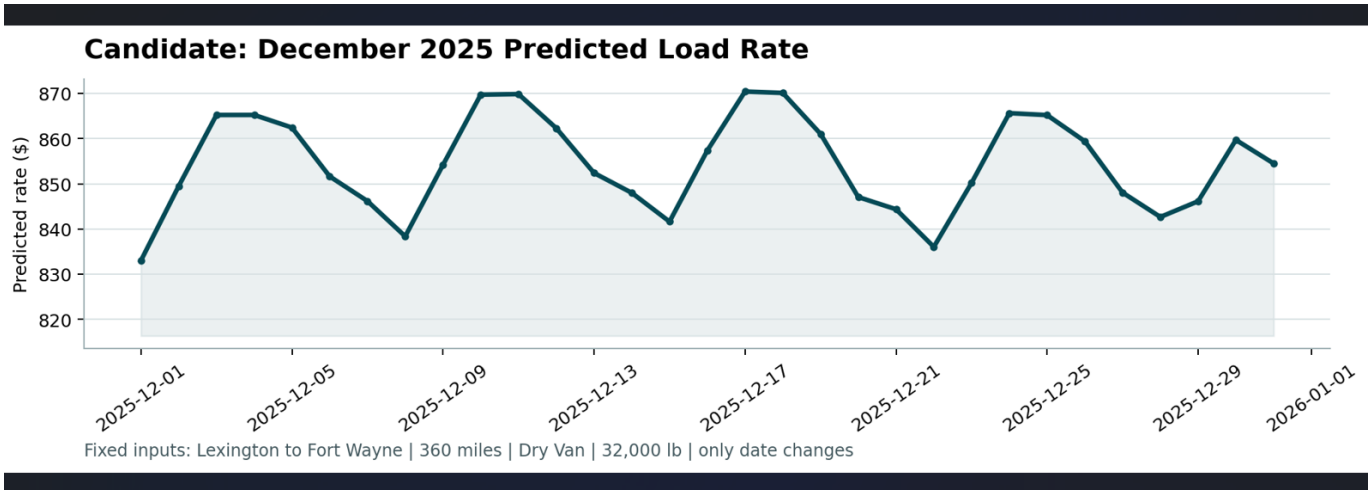
October 2025 chronological holdout results

Metric	Result
MAE	$117.01
RMSE	$647.14
MAPE	5.675%
R^2	0.8208
Selected iterations	466

These metrics describe the October chronological holdout only. They are not accuracy measurements for the unlabeled validation loads; final validation performance can only be calculated when ground-truth rates are available.

December 2025 fixed-route prediction chart

The supplied fixed inputs are Lexington to Fort Wayne, 360 miles, Dry Van, and 32,000 lb. Only the date changes across December 1–31, 2025. The supplied score.py completed its checks and generated the chart below.



Prediction output checks

The output file validation_predictions.csv contains exactly the required columns load_id,predicted_rate, with 12,000 rows, no missing predictions, no duplicate load IDs, and no non-positive predictions. The official scorer validated all 12,000 validation predictions and all 31 December predictions. Because validation labels are not supplied, the scorer does not report final validation accuracy.

Reproducibility

Install dependencies with *python -m pip install -r requirements.txt*, then run *python train_model.py --data-dir data --output-dir outputs*. Run the supplied scorer with the generated validation and December prediction files. Generated artifacts are written to outputs/ so the supplied source data and template remain unchanged.